BCH 216 — Biochemistry II

Practice Workbook & Answer Key | ABUAD 200 Level Pharmacy | Second Semester

An original revision aid by the EverythingABUAD student team | everythingabuad.com

How to use this workbook. The questions below are grouped by topic and were written from scratch by our team to help you revise the core concepts of BCH 216. They are original practice items for self-testing — not past exam papers and not a prediction of any future test. Try to answer each question on your own first, then check yourself against the answer key at the end. Always cross-check with your lecturer's current course outline.

Nucleic Acid Chemistry

1. Name the three components that make up a single nucleotide.

2. Which pentose sugars distinguish RNA from DNA?

3. Repeating nucleotides joined by phosphodiester bonds form what larger molecules?

4. Which scientist established that the ratio of purines to pyrimidines in DNA is always 1:1?

5. Classify the nitrogenous bases into their two structural families.

6. Which base is unique to RNA, replacing thymine found in DNA?

7. Through which ring atoms do the base and sugar attach in a nucleoside?

8. When heat-denatured DNA strands are cooled and re-pair, what is the process called?

9. In eukaryotes, DNA is packaged into long linear units known as what?

10. In which cell compartments is eukaryotic DNA primarily located?

11. In prokaryotes, the region holding the circular chromosome is called the ___.

12. What is the term for a nucleoside built on a ribose sugar with an adenine base?

13. Name the ribonucleoside formed from cytosine, and from uracil, respectively.

14. Give the abbreviation for the deoxyribonucleotide bearing cytosine in its diphosphate form.

15. RNA molecules with catalytic activity are known as ___.

16. What does it mean physically when DNA is described as being at its melting temperature (Tm)?

Central Dogma & Protein Synthesis

17. State the directional flow of genetic information described by the central dogma.

18. Name the three classes of RNA directly involved in making proteins.

19. Which DNA strand is read as the pattern for building mRNA?

20. Genetic information for a protein is carried in mRNA as a sequence of what units?

21. The genetic code is described by which two properties regarding codon-amino-acid matching?

22. What roles do tRNA molecules perform during translation?

23. Name the standard start codon and the three stop codons.

24. On which cellular structures does translation physically occur?

25. Synthesis of an mRNA copy from a DNA template is called what step?

26. Which enzyme builds DNA from an RNA template, reversing the usual flow?

27. In what year was the genetic code essentially fully deciphered?

DNA Replication

28. Replication is termed semiconservative. Explain what each daughter molecule contains.

29. Which enzyme unwinds the double helix at the replication fork?

30. The specific site where replication begins on the DNA is called the ___.

31. Which enzyme adds new nucleotides to the growing DNA strand?

32. DNA polymerase requires a short starter sequence to begin. What is it called?

33. Toward which end of the growing strand are new nucleotides added?

34. Contrast leading- and lagging-strand synthesis in one word each.

35. Which enzyme seals the gaps between adjacent DNA fragments?

Heme Degradation

36. In heme, the central iron is held inside which ring system, and by what bonding?

37. Which enzyme initiates heme breakdown by opening the porphyrin ring?

38. List the products released when heme oxygenase cleaves the heme ring.

39. Which enzyme converts biliverdin to bilirubin?

40. Why must bilirubin travel bound to a carrier in blood, and what is that carrier?

41. In the liver, bilirubin is conjugated with what to become water-soluble?

42. Name the gut-derived pigments produced from bilirubin in stool and urine.

43. Iron released from heme is stored in the body bound to which protein?

44. Accumulation of bilirubin in tissues produces which clinical sign?

45. State the approximate molecular size of heme oxygenase-1.

46. Which isoform of heme oxygenase is constitutively expressed (not stress-induced)?

Xenobiotic Metabolism

47. What is the chief aim of biotransformation of a foreign compound?

48. Name the principal Phase I oxidation reaction and the enzyme system that drives it.

49. Cytochrome P450 is named for the wavelength of its CO-bound absorption peak. State both.

50. Which cofactors supply reducing power to the P450 mono-oxygenase reaction?

51. Give four common Phase II conjugation reactions.

52. Phase II glucuronidation transfers glucuronic acid from which donor, and where?

53. Provide an example where biotransformation increases, rather than reduces, toxicity.

54. What term describes increased synthesis of metabolizing enzymes after drug exposure?

55. Name the tripeptide glutathione's three constituent amino acids.

56. Which antidote is used to treat acetaminophen (paracetamol) overdose?

57. What functional groups are typically introduced or exposed during Phase I metabolism?

58. Name the major P450 isoform responsible for metabolizing most clinical drugs.

59. Besides the liver, name two other organs capable of drug biotransformation.

60. What term describes drug loss during first transit through the gut wall and liver?

Biochemistry of Hormones

61. Define a hormone in functional terms.

62. Why do peptide hormones require a second messenger while steroids do not?

63. To what cellular structure do water-soluble peptide hormones bind, and what does this trigger?

64. Name the classic cyclic-nucleotide second messenger derived from ATP.

65. Which two second messengers arise from membrane phospholipid hydrolysis?

66. The steroid hormone-receptor complex binds which specific DNA sequence?

67. Which enzyme, activated via the G-protein alpha subunit, generates cAMP?

68. From which precursor molecule are all steroid hormones synthesized?

69. Give three examples of steroid hormones.

70. Glands that secrete hormones directly into the blood are termed ___; those using ducts are ___.

71. List three defining characteristics of hormone action.

72. Calcium acts as which class of intracellular messenger in some signalling cascades?

Genetics & Heredity

73. Define the terms genotype and phenotype.

74. How many chromosome sets are in a haploid cell, and how many chromosomes in a human haploid cell?

75. State the human chromosome complement: total pairs, autosome pairs, sex-chromosome pairs.

76. Define the terms homozygous and heterozygous.

77. The fixed position a gene occupies on a chromosome is called its ___.

78. Alternative forms of the same gene are known as ___.

79. Which proteins package DNA into chromatin?

80. Name the Rh-incompatibility condition affecting newborns.

DNA Repair & Mutations

81. Complete the relationship: all carcinogens are ___, but not all ___ are carcinogens.

82. What is the general term for agents that cause changes in DNA base sequence?

83. Mutations can disturb which three DNA-related processes?

84. Name the repair pathway that removes bulky, helix-distorting lesions such as UV dimers.

85. Which inherited disorder results from defective nucleotide excision repair?

86. State the typical inheritance pattern of xeroderma pigmentosum.

87. Which gene product recognizes and helps repair double-strand DNA damage?

88. Uncorrected DNA damage that escapes repair can ultimately lead to what disease state?

Recombinant DNA Technology

89. What is the broad term for deliberately altering an organism's DNA in the laboratory?

90. Which enzymes cut DNA at specific sequences, enabling cloning?

91. DNA fragments are joined into a vector using which enzyme?

92. Carrier molecules that ferry foreign DNA into a host cell are called ___.

93. Why are bacteriophages useful as cloning vectors for large inserts?

94. A collection of cloned DNA copies made from mRNA is called a ___.

95. Which enzyme synthesizes cDNA from an mRNA template?

96. Name the technique that uses an electric pulse to introduce DNA into cells.

97. Who invented the polymerase chain reaction, and in what year was the Nobel Prize awarded for it?

98. Which heat-stable polymerase makes PCR practical, and from what organism?

99. Name the three temperature steps of a PCR cycle (denature, anneal, extend).

100. Which instrument automatically cycles the temperatures during PCR?

101. State one limitation of using bacterial hosts to express human proteins.

102. By roughly what factor can PCR amplify a target sequence in a single run?

Answer Key

Worked answers, grouped to match the questions above.

Nucleic Acid Chemistry

1. A nitrogenous base, a pentose sugar, and a phosphate group.

2. RNA contains D-ribose; DNA contains 2-deoxy-D-ribose.

3. Polynucleotides (the backbone of nucleic acids).

4. Erwin Chargaff (Chargaff's rule).

5. Purines (adenine, guanine) and pyrimidines (cytosine, thymine, uracil).

6. Uracil replaces thymine; the difference is a methyl group on thymine.

7. N-9 of purines and N-1 of pyrimidines bond to C-1 of the pentose sugar.

8. Annealing (re-association / re-naturation).

9. Chromosomes.

10. The nucleus, with a smaller amount in the mitochondria.

11. Nucleoid.

12. Adenosine (the ribonucleoside of adenine).

13. Cytidine (cytosine) and uridine (uracil).

14. dCDP (deoxycytidine diphosphate).

15. Ribozymes.

16. Half of the double-helical structure has separated (denatured).

Central Dogma & Protein Synthesis

17. DNA -> RNA -> protein (replication, transcription, translation).

18. Messenger RNA (mRNA), ribosomal RNA (rRNA), and transfer RNA (tRNA).

19. The template (antisense) strand.

20. The genetic code, read in three-base codons.

21. Degenerate (several codons per amino acid) and unambiguous (one codon -> one amino acid).

22. They act as carriers, delivering amino acids to the ribosomes.

23. Start codon: AUG (codes for methionine). Stop codons: UGA, UAA, and UAG.

24. The ribosomes.

25. Transcription.

26. Reverse transcriptase (RNA-dependent DNA polymerase).

27. 1968.

DNA Replication

28. Each daughter DNA keeps one original (parental) strand and one newly synthesized strand.

29. Helicase (helicases break the hydrogen bonds between strands).

30. Origin of replication.

31. DNA polymerase.

32. A primer.

33. The 3' (three-prime) end.

34. Leading strand: continuous; lagging strand: discontinuous.

35. DNA ligase.

Heme Degradation

36. A porphyrin ring; iron is bound as a tetradentate ligand to four ring nitrogens.

37. Heme oxygenase (HO-1).

38. Biliverdin, carbon monoxide (CO), and ferrous iron, Fe(II).

39. Biliverdin reductase.

40. It is largely insoluble, so it travels complexed to serum albumin.

41. Glucuronate (glucuronic acid), forming bilirubin diglucuronide.

42. Stercobilin (faeces) and urobilin (urine).

43. Ferritin.

44. Jaundice.

45. About 32 kDa (around 288 amino acid residues).

46. HO-2.

Xenobiotic Metabolism

47. To convert a lipophilic, non-polar, often toxic substance into a more water-soluble, excretable one.

48. Hydroxylation, catalysed by the mono-oxygenases of the Cytochrome P450 system.

49. It absorbs maximally at 450 nm when bound to carbon monoxide (CO).

50. NADPH (and NADH).

51. Glucuronidation, acetylation, glutathione conjugation, and glycine conjugation.

52. From UDP-glucuronic acid, occurring in the microsomes.

53. Conversion of vinyl chloride to vinyl chloride epoxide, which binds DNA/RNA and is linked to liver cancer.

54. Induction.

55. Gamma-glutamic acid, cysteine, and glycine.

56. N-acetylcysteine.

57. Hydroxyl (-OH), amino (-NH2), and sulfhydryl (-SH) groups.

58. CYP3A4.

59. Any two of: gastrointestinal tract, lungs, skin, kidneys.

60. The first-pass effect.

Biochemistry of Hormones

61. A chemical messenger secreted in trace amounts by one tissue and carried by blood to stimulate a specific activity in a target tissue.

62. Peptide hormones are water-soluble and bind surface receptors; steroids are lipid-soluble and act on intracellular receptors.

63. Receptor molecules in the cell membrane, triggering production of a second messenger.

64. Cyclic AMP (cAMP).

65. Inositol triphosphate (IP3) and diacylglycerol (DAG).

66. The hormone response element (HRE).

67. Adenylate cyclase.

68. Cholesterol (a lipid).

69. Any three of: adrenal cortical hormones, androgens, estrogens.

70. Endocrine glands; exocrine glands.

71. Strict specificity of action, high biological activity at low concentration, and action remote from the site of release.

72. A second/third messenger (Ca^{2+} ions).

Genetics & Heredity

73. Genotype is the genetic constitution; phenotype is the observable expression of that genetic make-up.

74. One (1) set; 23 chromosomes in humans.

75. 23 pairs total: 22 pairs of autosomes and 1 pair of sex chromosomes.

76. Homozygous: identical alleles at a locus; heterozygous: two different alleles at a locus.

77. Locus (gene locus).

78. Alleles.

79. Histones.

80. Haemolytic disease of the newborn (HDN).

DNA Repair & Mutations

81. All carcinogens are mutagens, but not all mutagens are carcinogens.

82. Mutagens.

83. Base sequence integrity, replication, and DNA repair.

84. Nucleotide excision repair.

85. Xeroderma pigmentosum (XP).

86. Autosomal recessive.

87. The ATM gene product (recognition and repair of damaged DNA).

88. Cancer.

Recombinant DNA Technology

89. Genetic engineering.

90. Restriction endonucleases (nucleases).

91. DNA ligase.

92. Vectors (cloning / molecular vectors).

93. They can accommodate larger DNA inserts, suiting them to large genomic sequences.

94. cDNA (complementary DNA) library.

95. Reverse transcriptase (RNA-dependent DNA polymerase).

96. Electroporation.

97. Kary Mullis; the Nobel Prize in Chemistry was awarded in 1993.

98. Taq polymerase, from Thermus aquaticus.

99. Approximately 95 C (denaturation), 50 C (annealing), 72 C (extension).

100. The thermocycler.

101. Bacteria cannot perform eukaryotic post-translational modifications (plus ethical/regulatory concerns).

102. Up to about one million-fold (10^6).